

Sector Economics and Go-to-Market Viability

When a F1R3FLY shard pays for itself, and which markets get it there

A companion to “Rent and Shard Splitting”

F1R3FLY Industries — working note

June 11, 2026

A companion note. It reuses the cost model, the dense-packing result, and the rent lifecycle of “Rent and Shard Splitting” [1] and asks the next question: per market sector, how much monthly traffic does a shard need to be economically viable, which sectors produce it, and in what order should we deploy. All dollar figures are engineering estimates on stated assumptions; every fee is a lever, flagged as such, so the tables re-derive when real schedules replace ours.

1 The viability condition

A shard is viable when monthly phlogiston revenue covers its cost. From the rent note, one ten-node shard costs $C \approx \$4,160/\text{month}$ all-in, and dense elastic packing [2] lets it carry on the order of 30,000,000 fee-bearing transactions per month before the split threshold binds. If a sector’s transactions bear a blended fee r_{tx} (precharge, activation, and rent combined, in dollars), then break-even monthly traffic is simply

$$T^* = \frac{C_{\text{eff}}}{r_{\text{tx}}}, \quad C_{\text{eff}} = C \times (\text{compliance multiplier}).$$

Risk pushes on *both* sides of this ratio. A riskier sector can charge a higher r_{tx} , but it also raises C_{eff} : replication, audit, and dedicated infrastructure break the “all co-resident on one cloud” assumption. The useful dimensionless quantity is therefore not T^* but the *break-even load*, T^* divided by per-shard capacity—the fraction of a shard a sector must fill to pay for it.

Observation 1 (Capacity is never the gate; demand is.). *Every sector below is feasible on a single shard’s capacity—even the highest-volume case sits far under the split threshold per shard. Viability is gated entirely by whether a sector produces enough fee-bearing transactions, and the break-even load varies by more than two orders of magnitude across sectors. Riskier data is easier to make pay, not harder.*

Common parameters. Unless noted: base cost $C = \$4,160/\text{shard}/\text{month}$; per-payload network cost at dense packing $C/30,000,000 \approx \$0.00014$; low-risk shards run dense ($\sim 28,000,000$ – $30,000,000$ tx/mo), mission-critical shards run with headroom ($\sim 20,000,000$ tx/mo). Compliance multipliers: AI $1.2\times$, music $1\times$, biotech $1.5\times$, fintech $2.5\times$, medical $4\times$.

2 AI (ASI Alliance) — the growth engine

Assumptions: $r_{\text{tx}} = \$0.01$ (inference calls, tool invocations, and agent-to-agent settlement all bear real value); $C_{\text{eff}} = \$4,992$; $\sim 10,000$ tx per agent per month at machine pace; dense packing.

Break-even: $\sim 500,000$ tx/month = $\sim 1.8\%$ of a shard. An AI shard pays for itself at two percent utilisation.

Table AI. Agent population ramp at $10\times$ /year (unbounded).

Year	Agents	Monthly tx	Shards	Cost	Revenue	Margin
1	100	1.0M	1	\$4,992	\$10,000	$2.0\times$
2	1,000	10M	1	\$4,992	\$100,000	$20\times$
3	10,000	100M	4	\$19,968	\$1.0M	$50\times$
4	100,000	1.0B	36	\$179,712	\$10M	$56\times$
5	1,000,000	10B	358	\$1.79M	\$100M	$56\times$

The defining trait is the absence of a TAM ceiling. Machine-generated volume is unbounded and sticky; margin saturates near $56\times$ and then revenue simply scales. AI both load-tests the dispatch/COMM hot path under real volume *and* is the only sector whose growth does not terminate—which is why it should anchor the rollout rather than a low-fee consumer sector.

3 Fintech (Boring Financial) — the density engine

Assumptions: $r_{\text{tx}} = \$0.25$ blended (settlement, contract-lifecycle, and loan-servicing events); $C_{\text{eff}} = \$10,400$ (custody, audit, regulatory load); safety headroom.

Break-even: $\sim 42,000$ tx/month = $\sim 0.2\%$ of a shard—the most revenue-dense sector; a fintech shard pays for itself at one-fifth of one percent utilisation.

Table FIN. Transaction ramp.

Year	Monthly tx	Shards	Cost	Revenue	Margin
1	50k	1	\$10,400	\$12,500	$1.2\times$
2	500k	1	\$10,400	\$125,000	$12\times$
3	5M	1	\$10,400	\$1.25M	$120\times$
4	50M	3	\$31,200	\$12.5M	$400\times$
5	500M	25	\$260,000	\$125M	$481\times$

Remark 1 (The most assumption-sensitive number in the note.). $\$0.25/\text{tx}$ is right for high-value settlement events, where it is trivial against notional, and wrong by one to two orders of magnitude for high-frequency micropayments, where fintech behaves like AI on margins. The real schedule is basis-points-on-notional for large events and sub-cent for small ones, so fintech is bimodal; model the two instrument classes separately once Boring supplies their mix.

4 Medical (VA) — the cohort-capped annuity

The Veterans Health Administration serves roughly 9,100,000 enrolled veterans [5]; the 80% target is $\sim 7,280,000$ patients. Assumptions: $\tau = 50$ provenance-logged record accesses per patient per month; $R = \$1/\text{patient/month}$ in platform fees (a conservative 2–7% of full-EHR per-patient spend); $C_{\text{eff}} = \$16,640$; headroom.

Break-even: $\sim 16,640$ patients against a $\sim 400,000$ -patient shard = $\sim 4\%$ load. A medical shard pays for itself at four percent utilisation.

Table MED. 10×/year rollout against a finite cohort (saturates at year 4).

Year	Patients	Monthly tx	Shards	Cost	Revenue	Margin
1	7,280	0.36M	1	\$16,640	\$7,280	0.4× (loss)
2	72,800	3.6M	1	\$16,640	\$72,800	4.4×
3	728,000	36M	2	\$33,280	\$728,000	21.9×
4	7,280,000	364M	19	\$316,160	\$7,280,000	23.0×
5–10	7,280,000	364M	19	\$316,160	\$7,280,000	23.0× (flat)

Two facts dominate. First, the cohort is a hard wall: a literal 10×/year ramp saturates the finite population by year 4, so medical revenue is *logistic*—explosive then flat—and past saturation it is an annuity (~\$87M/year revenue on ~\$3.8M/year infrastructure), not a growth story. Cumulative decade revenue is ~\$620M, of which infrastructure is a ~4% rounding error. Second, the constraint is never economic: even tripling cost for high-availability replication (3××4× loading) leaves ~7–8× margin at cohort. The binding constraints are regulatory and operational.

Remark 2 (Risk sets the rent posture.). “COMMs *delivered while funded*” (*the rent note’s mortality semantics*) is a safe default for music but a safety failure for a patient-record access that silently never fires on fund depletion. Mission-critical residuals must be pinned to committed on-chain balance (*germ line*), never allowed to deplete to eviction. This is the strongest reason medical deploys last: the rent model itself is most stressed exactly where the data is least forgiving.

5 Music — the cold-tail fill

A stable of ten independent artists at ~1M non-overlapping followers, each audient streaming a conservative ten songs/month, is 100M plays/month. Per-play on-chain, that is ~4 shards (\$16,640/mo), and break-even is

$$r_{\text{tx}}^* = \frac{\$16,640}{100\text{M}} \approx \$0.000166 \text{ per play.}$$

Any network fee above ~1/60 of a cent per play is profitable; at \$0.0003/play the stable runs ~1.8×.

Table MUS. Margin on the ten-artist stable (100M plays/mo, 4 shards, \$16,640/mo).

Network fee/play	Monthly revenue	Cost	Margin
\$0.0001	\$10,000	\$16,640	0.6× (underwater)
\$0.000166	\$16,640	\$16,640	break-even
\$0.0003	\$30,000	\$16,640	1.8×
\$0.001	\$100,000	\$16,640	6.0×

Two design facts. Batching plays off-chain and settling (say 100:1) collapses 100M plays to 1M transactions on one shard at ~7× margin—strictly better economics *unless per-play provenance is the product*. If transparent per-play royalty and capability-attenuated rights are the differentiator over incumbents, we *want* the 100M on-chain transactions, and music becomes the abundant *cold-tail packing* that monetises the spare leaves of the AI and fintech shards (rent note, bid-rent stratification). Either way music is viable; the choice is provenance granularity versus margin. The

caveat is concentration: 100M/month resting on ten artists is roster-dependent and churns, which is why music is high-margin fill, never the anchor.

6 Web 2.0 — participatory advertising

Free-at-point-of-use consumer properties have no per-transaction user fee, so the framework above appears to fail—until advertising is made *participatory*. End users pay in tokenised impressions per data payload until their usage is paid off, then either switch to impression-free payloads or accumulate advertising tokens spendable in other markets. This both closes the monetisation gap and inverts the extraction relationship.

The mechanism rests on one ratio. A data payload costs \$0.00014 at dense packing; an ordinary display impression at a ~\$3 CPM [9] is worth \$0.003:

one ad impression covers the network cost of ~20 data payloads.

Because impressions are tokenised on-chain, each impression is itself a fee-bearing transaction that funds the payload it rides plus margin: the ad layer is self-funding volume, not overhead. When cumulative impressions exceed cumulative usage, the user crosses into the impression-free or net-earner fork.

Table WEB. Three Google properties; payloads as emails/sessions/edits. Volumes from Gmail [6], Maps [7], Workspace [8].

Property	Payloads/mo	Shards	Infra cost/mo	Break-even ads/user/day
Gmail (1,800,000,000 users)	~3.6T	~120,000	~\$500M	~3
Maps (2,000,000,000 users)	~3.0T	~100,000	~\$416M	~2–3
Docs (3,000,000,000 users)	~600B	~20,000	~\$83M	~0.3

Every property is funded if a user views two or three ads per day—far below real exposure—so the typical user crosses into accumulating spendable tokens. **That is the inversion:** in incumbent Web 2.0 the user is the product, their attention extracted and the surplus captured by the platform; here the user covers their own cost with attention and *keeps the surplus* as portable tokens. The network takes a thin per-payload margin at astronomical volume—a \$0.00002/payload take across Gmail’s 3.6T payloads is still ~\$72M/month. This is “the community is the moat” and the anti-enshittification thesis rendered as an income statement.

Observation 2 (The cross-market spend saves Docs.). *Docs has the lowest break-even (~0.3 ads/day) but the highest ad resistance—nobody wants ads in documents. The fork resolves it: users earn tokens on high-CPM, ad-tolerant surfaces (Maps’ local and sponsored context) and spend them on impression-free Docs. The cross-subsidy is carried in the user’s portable tokens, structurally unlike incumbent bundling, where the cross-subsidy is the platform’s lever over the user.*

Enter through a privacy wedge, not at Gmail scale. 120,000 shards and \$500M/month is a hyperscaler buildout; per-payload viability is trivial but absolute capital is enormous. The wedge is a privacy-premium slice: 1% of Gmail (~18,000,000 users who actively want data ownership) is ~36B payloads/month ≈1,200 shards ≈\$5M/month infrastructure, trivially funded—and “own your data, earn from your attention” self-selects exactly those adopters. Web 2.0 is a beachhead-then-scale play entered once the participatory rails are proven on the earlier sectors.

7 Acquisition and soft costs

Folklore puts incumbent direct-to-consumer acquisition at $\sim \$50/\text{head}$ (Facebook) and $\sim \$19/\text{head}$ (WhatsApp). Those are *retail* costs: paid per individual, one head at a time. F1R3FLY’s model is *wholesale*—court one supply-side anchor and acquire its entire downstream population in a single deal:

$$\text{CAC}_{\text{eff}} = \frac{A}{n c},$$

with A the cost to acquire one anchor (business development, onboarding, integration, contract setup), n the population that anchor brings, and c the fraction that converts. The denominator is what collapses the figure. A second difference matters more than the first: **the anchor is a paying customer, not a liability to be monetised later**. Because the artist’s sale embeds the transaction fee and the provider pays for the platform, acquisition is recovered from the anchor’s own fees—CAC is *self-liquidating*—and there is no consumer billing relationship with millions of end users, so collections, billing, and consumer-churn management are wholesale too.

Table CAC-1. Music 2014 court the artist, acquire the audience.

Scenario	A / artist	c	Audients/artist	CAC/head	vs. WhatsApp \$19
Inbound / high-convert	\$2,000	10%	100,000	\$0.02	$\sim 950\times$ cheaper
Mid	\$5,000	5%	50,000	\$0.10	$\sim 190\times$ cheaper
Conservative	\$10,000	1%	10,000	\$1.00	$\sim 19\times$ cheaper

Table CAC-2. Healthcare 2014 court the provider, own the patient.

Anchor	A	Patients	CAC/head	vs. Facebook \$50
Mid hospital system	\$500,000	1,000,000	\$0.50	$\sim 100\times$ cheaper
VA (conservative)	\$5,000,000	7,280,000	\$0.69	$\sim 72\times$ cheaper
VA (heavy gov. sale)	\$20,000,000	7,280,000	\$2.75	$\sim 18\times$ cheaper

Patient data ownership—the private key to a data wallet—does double duty: it is a close-rate multiplier on the provider sale (a consent, portability, and compliance differentiator that lowers effective A) and a downstream acquisition channel (a patient who owns their data carries it to other F1R3FLY services, seeding the rest of the portfolio at zero marginal CAC). The cross-market multiplier compounds this: with token portability, a head acquired in one market is a user in m markets, so blended CAC falls to CAC/m while LTV multiplies by m against that single acquisition. Music LTV is thin on streams alone ($\sim \$0.003/\text{head-month}$, a multi-year payback), but the same head transacting across artist direct-sales and participatory-ad surfaces pays back in weeks; healthcare needs no such rescue ($\sim \$1/\text{patient-month}$ over a multi-year relationship gives $\text{LTV}/\text{CAC} \sim 20\text{--}90\times$).

Remark 3 (The counter-forces are symmetric.). *Anchor acquisition is lumpy and front-loaded—the VA’s \$5–20M is spent years before the cohort ramps, a real cash-flow-timing and sales-cycle problem. Conversion c is genuinely uncertain and is the whole ballgame for music. And concentration cuts on the acquisition side too: lose an anchor and lose its entire population at once. Wholesale acquisition and wholesale churn are the same mechanism.*

8 The connector shard: monetising interoperability

Per-sector shards earn on work done *inside* them; the *connector* shard earns on value moving *between* them. A provider checks a box at deploy time to join, and F1R3FLY collects a fee on every cross-shard transaction. This is the interchange position (Visa/Mastercard, SWIFT, an internet exchange point), and three properties make it the profit centre.

First, the tax base grows superlinearly. If a shard transacts with counterparties roughly in proportion to who is reachable, the cross-shard fraction χ of its traffic rises toward $(n-1)/n$ as n shards connect:

Table CON-1. Cross-shard fraction under uniform mixing.

Connected shards n	2	5	10	50	100
$\chi \rightarrow (n-1)/n$	50%	80%	90%	98%	99%

Early on almost everything is intra-shard and the connector earns little; at scale nearly every transaction crosses a boundary and the connector taxes essentially the entire platform volume. Second, its cost stays tiny—it routes and clears coordination, it does not store payloads—so its margin is the highest in the system. Third, the opt-in box is a flywheel: each provider that joins makes the connector more valuable to every other member, raising the value of joining, and it captures *competitors* too, since any rival that wants interoperability connects and pays the toll. One need not win every sector; one needs every sector to connect.

Table CON-2. Connector revenue (illustrative; r_x and χ are levers).

Scenario	Connectable vol./mo	Effective χ	Fee r_x	Connector rev./mo
Core sectors, early	~11B	20%	\$0.005	~\$11M
Core sectors, dense network	~11B	60%	\$0.005	~\$33M
+ Web 2.0 connected	~7.2T payloads	5%	\$0.0005	~\$180M

Even the conservative core case rivals the largest single-sector line; once the mass-market substrate connects, the connector dwarfs everything at a fraction of the per-transaction fee. **Travel and hospitality** is the archetypal amplifier: a single trip is inherently multi-shard—payment, loyalty/identity, location, booking, reviews—so it carries a high native χ and lights up the connector far beyond its standalone volume. The correct valuation of any new sector is therefore *intra-shard fees plus its contribution to connector χ* .

This closes the loop with acquisition. The cheap wholesale CAC works *because* a head acquired in one market is a user in all of them, and the mechanism by which it is a user in all of them is cross-shard transactions—which are the connector’s revenue. Cross-market portability is simultaneously the CAC discount and the connector’s tax base: the thing that makes users cheap to acquire is the thing that makes them lucrative.

Remark 4 (Counter-forces.). *The connector is back-loaded like medical: with few shards connected χ is small and revenue is negligible for years, so the sectors must bootstrap volume first. Providers can bypass the toll via private peering (the peering-versus-transit dynamic of internet economics), which caps r_x ; because the box is opt-in, demand is elastic and revenue is maximised at an interior toll, not the highest one. A levy on all cross-sector flow is interchange, among the most heavily regulated prices in finance, so dominance carries a compliance tax. And concentration recurs: the connector’s value rests on the set of connected anchors.*

9 Bandwidth: the media exception

The cost model set egress to zero on the assumption that payloads fit inside OCI’s free tier. That is correct for KB-scale transactional payloads but wrong for media. OCI prices internet egress at \$0.0085/GB beyond the first 10 TB/month and inter-region transfer at \$0.005/GB [4] (Oracle reportedly moved to free egress across all regions in early 2026; we model the rate as a lever rather than baking in zero). Bandwidth cost is payload size $\times$ moves $\times$ rate, and per-payload it dwarfs the \$0.00014 compute/storage cost exactly when payloads are large:

Table BW. Per-payload egress versus the transactional cost floor.

Payload	Size	Egress @ \$0.0085/GB	@ CDN \$0.04/GB	Verdict
Fintech tx / AI token / EHR record	~1–5 KB	\$0.00000004	—	negligible
Audio, compressed	~4 MB	\$0.000034	\$0.00016	absorbable
Audio, lossless / hi-res	~40 MB	\$0.00034	\$0.0016	exceeds fee
Radiology study (CT/MRI)	~250 MB	\$0.002125	\$0.010	dominant cost

The verdict by sector follows the payload size. Bandwidth is **negligible** for the transactional core (AI, fintech, EHR metadata—the zero assumption holds). It is **material but absorbable** for medical imaging: at ~1.25 GB/patient-month the cohort moves ~9 PB/month, costing ~\$77k/month (~\$364k at CDN rates)—only ~1–5% of medical’s high per-patient revenue. It is **viability-threatening** only where a large payload meets a thin fee: lossless audio streaming costs ~\$0.00034/play in egress against a ~\$0.0003 streaming fee, so hi-res music is bandwidth-underwater and needs either a higher fee (~\$0.0005/play) or edge caching, whereas compressed audio (~11% of the fee) is merely a margin haircut.

Remark 5 (Move capabilities, not payloads.). *The OCap/data-wallet architecture mitigates this structurally. Cross-shard and through the connector, the system transmits an access token (KB), not the blob (GB); the large object lives once, content-addressed, in the patient’s data wallet or an edge cache near where it is consumed, and bandwidth is incurred only at the final serve hop. Three consequences: the connector economics are unaffected (cross-shard traffic is tokens, so the profit centre carries no bandwidth); bandwidth becomes a per-serve, edge-localised cost that should be metered in phlogiston as a third resource—a per-byte egress pass-through alongside rent (standby) and activation (COMM)—so that whoever moves the blob pays for the bytes; and edge caching of hot blobs (radiology priors, popular tracks) amortises egress over serves, which is what rescues lossless audio. On OCI specifically, cross-data-centre movement is cheap (inter-region \$0.005/GB, and FastConnect to on-premises charges a flat port fee with no per-GB cost), so the expensive hop is not between data centres but the last mile to the consumer.*

10 Synthesis and rollout

The pattern that falls out: **revenue density and risk rise together, and both run opposite to volume.** The original safe-to-critical *risk* ordering is correct and unchanged. Layered on top, the *economic* ordering is:

1. **AI — anchor.** Simultaneous load-testing and viability; hardens the hot path; unbounded growth.

Table SUM. The five sectors plus Web 2.0.

Sector	Revenue density	Risk	Boundedness	Break-even load	Role
Web 2.0	ad-funded (\$0.00014/pl)	low-mod	astronomical	~3 ads/user/day	mass-market moat (last)
Music	\$0.0003/tx	low	unbounded, churny	~near full	cold-tail fill
AI	\$0.01/tx	low-mod	unbounded, sticky	~2%	anchor / growth
Fintech	~\$0.25/tx	high	semi-bounded	~0.2%	margin engine
Medical	high/patient	highest	hard cohort cap	~4%	cash-cow annuity

2. **Fintech** — **margin**. Highest revenue density rewards even modest volume; builds high-assurance discipline on financially recoverable failures.
3. **Medical** — **annuity**. Highest margin per unit, capped at the cohort, deployed last so the assurance and germ-line-pinning machinery is battle-tested first.
4. **Music** — **continuous fill**. Not a phase but the cold-tail packing that monetises spare capacity across the above.
5. **Web 2.0** — **mass-market substrate**. Entered through a privacy wedge once the participatory ad rails are proven.

11 Sensitivity and open levers

Every figure scales transparently. The fees r_{tx} are the dominant levers and should be replaced with real schedules from ASI and Boring as soon as available; the fintech number in particular is bimodal and warrants two instrument classes. Acquisition adds three more levers (A , conversion c , population per anchor) and the connector adds two (r_x , effective χ); the connector’s revenue is the most sensitive of all to χ , which is itself a function of how many shards connect. The per-payload cost falls linearly with packing density, so the entire Web 2.0 case strengthens with the elastic-hashing gains the rent note quantifies—for low-fee consumer volume, dense packing is the difference between profit and loss, not a refinement. The compliance multipliers are coarse stand-ins for replication, audit, and dedicated-infrastructure costs and should be replaced with per-sector infrastructure models before any commitment. Finally, the single largest non-modelled risk is customer concentration—ten artists in music, one federal buyer in medical—which the portfolio sequencing above is in part designed to diversify.

References

- [1] F1R3FLY Industries. Rent and shard splitting: a storage-economics companion to occupancy-driven sharding. Working note, 2026.
- [2] M. Farach-Colton, A. Krapivin, and W. Kuszmaul. Optimal bounds for open addressing without reordering. arXiv:2501.02305, 2025.
- [3] L. G. Meredith and M. Radestock. A reflective higher-order calculus. *Electronic Notes in Theoretical Computer Science*, 141(5):49–67, 2005.

- [4] Oracle Corporation. Oracle Cloud Infrastructure pricing (compute, block volume, and networking price lists). <https://www.oracle.com/cloud/price-list/>, accessed June 2026.
- [5] U.S. Department of Veterans Affairs, Veterans Health Administration. About VHA: enrolled-veteran population and budget. <https://www.va.gov/health/aboutvha.asp>, accessed June 2026.
- [6] Gmail usage statistics: active users and daily message volume, 2026. Aggregated industry reporting (Statista, Litmus, Radicati), accessed June 2026.
- [7] Google Maps usage statistics: monthly active users, 2026. Aggregated industry reporting, accessed June 2026.
- [8] Google Workspace active-user disclosures, 2026. Google I/O and aggregated industry reporting, accessed June 2026.
- [9] Display advertising benchmarks 2026: Google Display Network average CPM $\approx$ \$3.12. Aggregated programmatic-benchmark reporting, accessed June 2026.